

CORPORATE SECURITY & DISASTER RECOVERY FRAMEWORK

IT Support Professional
Certificate Capstone Project

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Introduction

Ubuntu Innovations (Pty) Ltd relies heavily on information technology systems, cloud services, digital communications, and business-critical data to support daily operations. As cyber threats continue to evolve, organisations must implement effective security controls, disaster recovery strategies, and employee awareness programs to reduce operational risk and maintain business continuity.

This document presents the proposed Backup and Disaster Recovery Plan, Corporate Cybersecurity Framework, Incident Response Plan, and Security Awareness Training Program for Ubuntu Innovations (Pty) Ltd. These measures are designed to protect company assets, support secure operations, and ensure the organisation can respond effectively to security incidents and unexpected disruptions.

Backup and Recovery Plan

Introduction

Ubuntu Innovations (Pty) Ltd is a growing technology company based in Cape Town with 25 employees working across five key departments: Executive Management, Finance, Human Resources, Sales and Marketing, and Software Development.

As a technology-focused organisation, Ubuntu Innovations relies heavily on digital information to conduct its daily operations. Critical business data includes financial records, employee information, customer relationship data, software development projects, source code repositories, marketing content, contracts, business reports, and operational documentation.

The loss, corruption, or unavailability of this information could significantly impact the organisation's ability to operate effectively. Potential threats include hardware failure, ransomware attacks, accidental file deletion, power outages, system misconfiguration, theft of equipment, and natural disasters such as fire or flooding.

This Backup and Disaster Recovery Plan establishes procedures, technologies, responsibilities, and recovery objectives required to ensure business continuity and protect company data. The plan is designed to minimise downtime, reduce data loss, and allow Ubuntu Innovations to recover quickly from unexpected incidents.

Purpose of the Backup and Disaster Recovery Plan

The purpose of this plan is to:

- Protect critical business data from loss or corruption.
- Ensure business operations can continue following a disruption.
- Reduce the financial impact of system failures and cyber incidents.
- Support compliance with company policies and data protection requirements.
- Provide a structured approach for restoring systems and information.
- Protect the intellectual property developed by the Software Development Department.
- Safeguard sensitive employee information managed by the Human Resources Department.

- Preserve financial records required for accounting, payroll processing, and auditing purposes.

Without a proper backup and recovery strategy, Ubuntu Innovations could experience significant operational delays, financial losses, reputational damage, and loss of customer confidence.

Backup Infrastructure

To support reliable data protection, Ubuntu Innovations will implement a hybrid backup infrastructure consisting of:

Network Attached Storage (NAS)

The company will utilise the recommended 8TB RAID-enabled NAS device identified in the Week 1 Hardware Inventory.

The NAS device will serve as the primary onsite backup repository.

Benefits include:

- Centralised storage management.
- Fast backup and recovery performance.
- RAID redundancy for improved fault tolerance.
- Secure access control.
- Support for automated backup scheduling.

Cloud Backup Storage

In addition to local backups, Ubuntu Innovations will maintain secure offsite cloud backups using Microsoft Azure Backup or Microsoft OneDrive for Business.

Cloud storage provides an additional layer of protection by ensuring backup copies remain available even if the physical office experiences a disaster.

Benefits include:

- Geographic redundancy.
- Protection against theft and fire.
- Secure encrypted storage.
- Remote accessibility.
- Improved disaster recovery capabilities.

Data Classification and Backup Scope

Not all company data has the same level of importance. The following information assets have been identified as critical and must be included in backup operations.

Finance Department

Data to be protected:

- Payroll information.
- Tax documentation.
- Budget reports.
- Financial statements.
- Sage Accounting data.

Importance:

Loss of financial information could disrupt salary payments, tax submissions, budgeting activities, and financial reporting obligations.

Human Resources Department

Data to be protected:

- Employee records.
- Recruitment documentation.
- Leave records.
- Performance evaluations.
- Employment contracts.

Importance:

HR information contains confidential employee data that must be protected from loss and unauthorised access.

Sales and Marketing Department

Data to be protected:

- Customer relationship data.
- Marketing campaigns.
- Sales reports.
- HubSpot CRM records.
- Digital marketing content.

Importance:

Loss of customer information could affect revenue generation and damage customer relationships.

Software Development Department

Data to be protected:

- Source code.
- Project documentation.
- Databases.
- Testing environments.
- Configuration files.
- GitHub repositories.

Importance:

Software development assets represent valuable intellectual property. Their loss could delay projects and impact customer commitments.

Executive Management

Data to be protected:

- Strategic plans.
- Contracts.
- Business reports.
- Legal documentation.
- Board meeting records.

Importance:

These records support organisational decision-making and legal compliance.

Backup Strategy

Ubuntu Innovations will adopt the 3-2-1 Backup Rule.

The 3-2-1 rule requires:

- Three copies of important data.
- Two different storage locations.
- One copy stored offsite.

This approach significantly reduces the likelihood of permanent data loss.

Daily Incremental Backups

Daily incremental backups will occur automatically every evening at 20:00.

What Is an Incremental Backup?

An incremental backup only copies files that have changed since the previous backup.

Why It Is Important

Because Ubuntu Innovations operates daily business processes, new files are constantly being created and modified.

Examples include:

- New software code commits.
- Updated customer records.
- Employee leave applications.
- Financial transactions.

Incremental backups ensure that recent changes are protected without consuming excessive storage space.

Implementation

- Automated using backup software.
- Stored on the NAS device.
- Monitored by IT administrators.
- Backup logs reviewed daily.

Weekly Full Backups

Every Saturday evening a complete backup of all company systems will be performed.

What Is a Full Backup?

A full backup creates a complete copy of all selected data regardless of previous backups.

Why It Is Important

Full backups provide a reliable recovery point and simplify restoration procedures.

If multiple incremental backups become corrupted, the company can restore from the most recent full backup.

Implementation

The backup will include:

- Departmental folders.
- User profiles.
- Databases.
- Configuration settings.

- Shared storage locations.

Copies will be stored on both the NAS device and cloud backup platform.

Monthly Archive Backups

On the first day of each month, an archive backup will be created.

Purpose

Archive backups preserve historical information for long-term retention and compliance purposes.

Examples include:

- Financial reports.
- Payroll records.
- HR documentation.
- Completed development projects.
- Business contracts.

Retention Schedule

Monthly archives: 12 months

Annual archives: 7 years

Implementation

Archive copies will be stored in secure cloud storage and encrypted external storage devices.

Offsite Cloud Storage

Offsite storage is one of the most important elements of a disaster recovery strategy. If the company's office experiences:

- Fire
- Flooding
- Theft
- Electrical damage
- Physical sabotage

Local backups stored only on the NAS device may also be destroyed.

For this reason, Ubuntu Innovations will replicate all weekly and monthly backups to cloud storage.

Security Measures

Cloud backups will be protected through:

- AES-256 encryption.
- Multi-factor authentication.
- Role-based access control.
- Activity logging and monitoring.
- Secure HTTPS data transmission.

Recovery Objectives

Recovery objectives define how quickly systems must be restored and how much data loss is acceptable.

Recovery Time Objective (RTO)

The Recovery Time Objective represents the maximum acceptable amount of downtime.

Target RTO: 4 Hours

This means that critical services should be restored within four hours after a major incident.

Examples:

- File server outage.
- NAS device failure.
- Ransomware attack.
- Hardware failure.

A four-hour target is appropriate because Ubuntu Innovations depends heavily on technology for daily operations.

Recovery Point Objective (RPO)

The Recovery Point Objective represents the maximum acceptable amount of data loss.

Target RPO: 24 Hours

Since incremental backups occur daily, the organisation should not lose more than one business day's worth of information.

This ensures that customer records, development work, and operational documents remain largely protected.

Disaster Recovery Procedures

Scenario 1: NAS Device Failure

1. Identify failure through monitoring alerts.
2. Replace faulty hardware.
3. Restore latest backup from cloud storage.
4. Verify data integrity.
5. Resume operations.

Scenario 2: Ransomware Attack

1. Disconnect affected systems immediately.
2. Notify IT support personnel.
3. Identify infected devices.
4. Remove malicious software.
5. Restore clean backup copies.
6. Verify system functionality.
7. Return systems to production.

Scenario 3: Office Disaster

Examples:

- Fire
- Flood
- Theft
- Extended power outage

Recovery Process:

1. Obtain replacement hardware.
2. Access cloud backup platform.
3. Restore critical business systems.
4. Re-establish network services.
5. Validate restored data.
6. Resume business operations.

Backup Testing and Maintenance

Backups are only useful if they can be successfully restored. Ubuntu Innovations will conduct quarterly testing to verify backup integrity.

Testing activities include:

- File restoration testing.
- Database restoration testing.
- Full system recovery simulations.
- Cloud backup verification.
- Disaster recovery exercises.

Results will be documented and reviewed by IT management.

Conclusion

The Backup and Disaster Recovery Plan provides Ubuntu Innovations (Pty) Ltd with a comprehensive strategy for protecting business-critical information and maintaining operational continuity. Through the implementation of daily incremental backups, weekly full backups, monthly archive backups, secure offsite cloud storage, and clearly defined recovery objectives, the organisation can significantly reduce the risk of data loss and minimise downtime following a disruptive event.

The plan supports the needs of every department while ensuring that customer data, employee information, financial records, and software development assets remain secure, recoverable, and available when needed.

Corporate Cybersecurity Framework

Cybersecurity Policy

This policy establishes the minimum security standards for all company information systems, devices, and personnel. Compliance is mandatory to safeguard organization data and infrastructure against evolving cyber threats.

Password Complexity Requirements

To prevent brute-force attacks and unauthorized access, all user accounts must adhere to strict complexity standards:

- **Minimum Length:** Passwords must be at least 14 characters long.
- **Character Variety:** Must include at least three of the following four categories: Uppercase letters (A-Z), Lowercase letters (a-z), Numbers (0-9), and Special characters (e.g., !, @, #, \$, %, ^, *).
- **Expiration & Reuse:** Passwords do not expire automatically unless a breach is suspected (aligned with modern NIST guidelines), but the system will block the reuse of the last 5 history passwords.
- **Prohibited Items:** Passwords cannot contain the user's account name, real name, or easily guessable dictionary words (e.g., 'Password123').

Multi-Factor Authentication (MFA)

MFA adds a critical layer of defence beyond standard credentials.

- **Mandatory Enforcement:** MFA is required for all employees accessing corporate systems, including email, VPNs, cloud applications (e.g., Microsoft 365, AWS), and HR portals.
- **Approved Factors:** Authentication must utilize an approved authenticator app (e.g., Microsoft Authenticator, Google Authenticator) via push notifications or Time-based One-Time Passwords (TOTP).
- **Disallowed Methods:** SMS and voice-call authentication are strictly prohibited due to SIM-swapping vulnerabilities, unless explicitly approved by the IT Security team as a temporary fallback.

Device Encryption

Data-at-rest must be protected against physical theft or loss of hardware.

- **Full Disk Encryption (FDE):** All corporate-issued endpoints (laptops, desktops, smartphones) must have full disk encryption enabled.

- Windows Devices: Must utilize BitLocker with a minimum of AES-256 bit encryption.
- macOS Devices: Must utilize FileVault 2.
- Key Management: Encryption recovery keys must be securely escrowed and managed centrally by the IT department; users are prohibited from modifying encryption settings.

Antivirus & Endpoint Detection and Response (EDR) Standards

Continuous monitoring and malware prevention must be maintained on all endpoints.

- Deployment: Every endpoint must run the centralized corporate EDR/Antivirus solution (e.g., Microsoft Defender for Endpoint).
- Configuration: Real-time scanning and behavioral analysis must remain active at all times. Definition updates must occur automatically at least once every 24 hours.
- Tamper Protection: Users are strictly barred from disabling, pausing, or uninstalling security software. Any alert generated will automatically log to the central Security Operations Center (SOC).

Patch Management

Software vulnerabilities must be mitigated swiftly to prevent exploitation.

- Automated Updates: Operating systems and third-party applications (browsers, productivity suites) must be enrolled in automated patch management cycles.
- Deployment Timelines: Patches will be deployed based on severity rankings:
 - Critical/Zero-Day Vulnerabilities: Patched within 48 hours of release.
 - High Severity: Patched within 14 days of release.
 - Medium/Low Severity: Patched within the standard monthly maintenance window (30 days).
- Reboot Policy: If a patch requires a system restart, users will receive a maximum 72-hour deferral window before a forced reboot occurs.

Acceptable Use Policy (AUP)

Employees must utilize corporate technology resources responsibly, ethically, and legally.

- Business Resource Only: Company systems, networks, and devices are provided primarily for business purposes. Incidental personal use is

permitted if it does not interfere with productivity or compromise security.

- **Prohibited Activities:** Employees shall not use corporate assets to access, download, or distribute pornographic, hate speech, or illegal material; engage in gambling, torrenting, or unauthorized file-sharing; or install unapproved third-party software or shadow IT applications.
- **Reporting Obligation:** Employees must immediately report any suspected security incidents, lost devices, or phishing attempts to the IT Helpdesk.

Risk Assessment Matrix

The following matrix identifies core operational risks, assesses their likelihood and impact, and outlines specific mitigation controls, informed by standards from Microsoft Learn, CompTIA Security+, and Google IT Support Frameworks.

Risk	Likelihood	Impact	Mitigation Controls
Phishing	High	High	Ongoing security awareness training; annual phishing simulations; robust email filtering gateways (DMARC/DKIM/SPF enforcement).
Ransomware	Medium	High	Implementation of localized EDR; strict network segmentation; automated, immutable, and offsite (3-2-1 rule) backup strategies.
Hardware Failure	Medium	Medium	Server and storage redundancy (RAID configurations); maintaining on-site spare hardware; documented hardware lifecycle replacement plans.
Power Outage	Medium	High	Uninterruptible Power Supplies (UPS) for localized servers/networking gear; automated backup generators for critical data centres; migration of core infrastructure to the cloud.
Data Leakage (Insider Threat)	Medium	High	Implementation of Data Loss Prevention (DLP) software; strict role-based access control (RBAC);

			principle of least privilege enforcement.
Unpatched Vulnerabilities	High	Medium	Routine vulnerability scanning; strict adherence to the patch management policy timelines; automated asset management discovery.

Incident Response Plan (IRP)

This plan provides a structured, step-by-step approach to managing security incidents efficiently to limit damage and reduce recovery time.

1. Detection

The goal is to identify a potential security incident as early as possible.

Indicators of Compromise (IoCs): Look for unusual network traffic spikes, unauthorized access logs, repetitive failed login attempts, or unexpected system configurations.

Alerting Channels: Automated alerts from Endpoint Detection and Response (EDR) systems, Firewalls, or Security Information and Event Management (SIEM) tools.

Verification: The Incident Response Team (IRT) quickly assesses the alert to confirm if it is a true security incident or a false positive.

2. Reporting

Clear communication channels ensure that the right people are notified without creating unnecessary panic.

Internal Escalation: Employees report anomalies to the IT/Security Helpdesk. High-severity events are immediately escalated to the Incident Commander.

External Notification: If sensitive data or Personally Identifiable Information (PII) is compromised, the Legal and Compliance teams determine reporting timelines for regulatory bodies (e.g., GDPR, HIPAA, or local data protection authorities) and affected customers.

Documentation: Log all details in a centralized incident ticketing system (Time of detection, systems involved, observed behaviour).

3. Containment

The primary objective here is to "stop the bleeding" and prevent the threat from spreading further into the network.

Short-Term Containment: Isolate affected devices from the local network and internet (e.g., disconnect network cables or place endpoints in an isolated VLAN).

Long-Term Containment: Apply temporary security patches, disable compromised user accounts, and close specific firewall ports being utilized by the attacker.

4. Eradication

Once the threat is contained, you must completely remove the root cause of the incident from the environment.

Threat Removal: Delete malware, malicious artifacts, and rogue registry keys.

Vulnerability Remediation: Patch the initial entry point (e.g., update vulnerable software or fix misconfigured cloud buckets).

Identity Reset: Force a global password reset for all compromised or highly privileged accounts.

5. Recovery

This phase focuses on safely restoring systems back to normal production operations.

System Restoration: Restore systems from clean, verified backups.

Validation: Continuous monitoring of restored systems for a set period (e.g., 48–72 hours) to ensure the threat does not resurface.

Phased Reconnection: Gradually bring systems back online to closely monitor network behaviour.

6. Lessons Learned

Often overlooked, this is the most critical phase for improving long-term security posture.

Post-Incident Review: Hold a meeting with the IRT and stakeholders within 1 to 2 weeks of the incident.

Key Questions: What went well? Where did the plan fail? How can we prevent this exact type of incident from happening again?

Plan Updates: Update this Incident Response Plan and adjust security controls based on the findings.

Security Awareness Training Outline

Target Audience: All Employees

Module 1: Phishing Awareness

The Concept: Explaining how attackers use deceptive emails, texts (Smishing), or phone calls (Vishing) to steal credentials or install malware.

Red Flags: Urgency/fear tactics, generic greetings, mismatched sender email domains, and suspicious attachments/links.

Action Items: Hover over links before clicking; never enter credentials on unverified pages; use the internal "Report Phishing" button.

Module 2: Password Security & Identity Protection

Password Strength: Moving away from short, complex passwords to long, memorable Passphrases (e.g., Correct-Horse-Battery-Staple).

The Golden Rule: Never reuse passwords across corporate and personal accounts.

Multi-Factor Authentication (MFA): How MFA acts as the ultimate second line of defence, and how to handle MFA push-notification fatigue (never approve a prompt you didn't trigger).

Module 3: Safe Internet Usage

Secure Connections: Understanding the difference between HTTP and HTTPS. Avoid using public Wi-Fi networks without an approved corporate VPN.

Shadow IT: The dangers of downloading unauthorized software or uploading sensitive corporate data into unvetted public AI tools or cloud storage.

Social Engineering on the Web: Recognizing fake update pop-ups or malicious search engine results.

Module 4: Device & Physical Security

Lock Before You Walk: Emphasizing the habit of locking screens (Win + L or Cmd + Ctrl + Q) every time an employee leaves their desk.

Physical Theft Prevention: Safe handling of company laptops in public spaces (cars, coffee shops, airports).

Removable Media: A strict policy against plugging unknown USB drives into company machines ("Don't pick up USB sticks in the parking lot").

Module 5: Incident Reporting (What to do when things go wrong)

No-Blame Culture: Cultivating an environment where employees aren't afraid to admit they clicked a bad link. Early reporting saves data.

How to Report: Clear instructions on how to reach the Security/IT team (Specific email extension, internal phone extension, or ticketing portal).

Immediate Action: If a screen locks up with a ransom note or starts behaving erratically, disconnect the network immediately and call IT.

Conclusion

The implementation of a comprehensive Backup and Disaster Recovery Plan, Corporate Cybersecurity Framework, Incident Response Plan, and Security Awareness Training Program will strengthen Ubuntu Innovations (Pty) Ltd's overall security posture and operational resilience.

By combining preventative security controls, structured response procedures, employee awareness initiatives, and reliable backup strategies, the organisation can significantly reduce the impact of cyber threats, system failures, and business disruptions. These measures support business continuity, protect critical information assets, and ensure that Ubuntu Innovations remains secure, productive, and prepared for future challenges.
